

Federated Drift Compensation via Active Learning in Edge Deployed Neural Networks

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Abstract—Edge-deployed neural networks frequently operate in an evolving physical domain, where slow and subtle shifts in the environment alter the properties of the sensor stream statistics. Thus, distributions of incoming observations change with time, which significantly impacts the prediction accuracy of models trained only on historical data. While centralized model retraining procedures address this issue, they require large amounts of data for retraining, which contradicts the data-minimization principles and the available bandwidth on edge devices. In order to resolve the stated problem, the paper presents an active learning based drift compensation framework that is localized and relies on the acquisition function of Bayesian Active Learning by Disagreement (BALD). The latter selects informative training samples from the out-of-distribution domain with limited annotation needs, thus, reducing the amount of unstructured data that must be processed to update the model. The framework synchronizes the locally acquired information through a private federated mechanism. The extensive simulations of environment brightness decay, demography drift, and physical occlusions performed on the five edge nodes confirm the effectiveness of the proposed approach. In particular, our active learning sample selection mechanism managed to recover 79.6% of the observed temporal degradation with only about 15% of all video frames annotated. Furthermore, this study demonstrates that strict layer-freezing geometries are not merely communication optimizations, but are mathematically required constraints to ensure learning trajectory stability under the obfuscation noise inherent to privacy-preserving federated aggregations.

Keywords—Federated Drift Compensation, Active Learning Stability, Bayesian Uncertainty, Epistemic Disagreement, Sparse-Label Adaptation, Non-IID Stabilization.

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INTRODUCTION

The dominant thesis of this paper is that epistemic-uncertainty-guided sample acquisition enables sustainable federated drift adaptation under sparse-label constraints. The Edge AI systems are inherently developed to operate in a self-sustaining environment that is constantly changing [Ref]. Contrary to the controlled conditions under which standard performance tests are performed, the characteristics of the environment at the Edge change dynamically, leading to shifts in the statistics of the data acquired by sensors and consequent degradation of performance of analytical or predictive models, also known as concept drift [Ref].

The Drift Adaptation Problem Under Label Scarcity

Federated synchronization alone does not provide fresh supervision. When an edge node encounters a drifted distribution, it lacks ground-truth annotations to orient the optimization trajectory. Static edge models become increasingly detached from the evolving environment, and naive continuous annotation is infeasible due to the resulting annotation bottlenecks created by randomly requesting human verification on a continuous stream [Ref]. The key issue is not distributed optimization but rather selecting when to acquire

supervision.

In practical deployments, temporal distribution shifts manifest across three operationally distinct vectors:

- High-Frequency Covariate Shift (Lighting/Weather): Daily or seasonal changes in illumination can cause significant distortions in the coordinate distribution. Such distortions reduce accuracy by up to 10–15% over a period of six months [Ref]. The covariate drift mostly affects geometrical characteristics of the feature space, not the class-separability properties.
- Low-Frequency Prior Probability Shift (Demographics): Gradual population turnover and changing behavioral regularities will skew underlying categorical representations. In the former case, this skew is different from mere feature drift: the model encounters fundamentally new class configurations.
- Structural Hardware Degradation (Occlusion/Focal): Lens scratching, mounting shifts, and thermal degradation corrupt bounding box topologies and spatial geometries [Ref]. Hardware degradation corrupts abstraction quality rather than semantic distributions directly.

Contributions and Subsystem Scope

This research proposes an epistemic uncertainty-aware adaptation framework that treats local drift mitigation specifically as a resource-constrained active data acquisition problem. The primary contributions are:

- We formulate temporal adaptation using Bayesian Active Learning by Disagreement (BALD) executed via Monte Carlo dropout, separating epistemic uncertainty (model ignorance) from aleatoric noise (environmental blur) directly on the edge node.
- We empirically demonstrate that strict layer-freezing geometries are not merely communication optimizations but stability-preservation constraints required under privacy-noise amplification.
- We show that selective sampling can restore up to 79.6% of drift-induced degradation via highly compressed labeling requests.

Within the ScholarMaster architecture, this subsystem performs uncertainty-guided drift adaptation, sparse-label acquisition, and federated adaptation stabilization only. It does not optimize hierarchical aggregation (Paper 14), synthesize deployment architecture (Paper 20), validate runtime enforcement (Paper 18), derive formal verification theorems (Paper 21), or define irreversible privacy doctrine (Paper 17).

RELATED WORK

Concept Drift in Streaming Environments

The machine learning community has extensively studied environmental drift, primarily grouping methodologies into passive and active adaptation suites. Passive adaptation ensembles, such as online gradient tracking, continuously update parameters in real-time [Ref]. On the other hand, trigger-based adaptation necessitates formal testing to identify the thresholds of the distribution shift, e.g., by means of a Page-Hinkley test [Ref]. Temporal methods indeed are well-established, but mostly rely on availability of ground-truth labels at any time-instant upon request, which is not always the case in an uncontrolled environment.

Bayesian Uncertainty and Active Learning

Our work fits within the literature of active learning, where an agent queries an oracle (in our case, human input) for information on only the most informative examples. Settles gives a broad overview of the field, illustrating how query strategies which utilize entropy of class probabilities produce much better results than simple random sampling [Ref]. While the classic softmax entropy captures overlapping confidence regions, Gal et al. were first to apply MC dropout to define a practical variational inference scheme of Bayesian Neural Networks, where one can calculate the mutual information and separate knowledge and noise induced entropy [Ref], [Ref]. Our contribution is to ground MC dropout within a distributed sensing

constraint topology.

Distributed Optimization Stability

Federated convergence bounds strongly depend on the variance among participating client updates. Zhao et al. theoretically established that highly divergent (non-IID) local objective trajectories severely retard global progression [Ref]. Furthermore, integrating differential privacy mechanics intrinsically injects scalar noise relative to parameter dimensionality [Ref]. Prior studies typically approach this as a trade-off, but fundamentally fail to explore the destructive feedback loops that occur when sparse active labels combine directly with high-dimensional noise injections—a gap this paper resolves via strict geometric layer-freezing bounds.

PROBLEM FORMULATION & DATA ABSTRACTION

To orient the active learning algorithms correctly, we formally define the boundary limits of localized distribution drift over time.

Temporal Non-IID Evolution

Let the overarching global dataset space be abstracted into finite partitions managed by N edge devices. The localized distribution at physical node i observed during time window t_1 is defined as $P_{i}^{(t_1)}(x,y)$. Under ideal static conditions, the test distribution perfectly mimics the training space. However, we define the environment as non-stationary.

We express the degree of inter-temporal drift at a single location using the Wasserstein metric W_1 . The distribution diverges significantly over a time delay Δt such that:

Equation: $W_1(P_{i}^{(t_1)}, P_{i}^{(t_1 + \Delta t)}) = \inf_{\gamma \in \Gamma(P, Q)} \int_{\mathcal{X} \times \mathcal{Y}} \|z - z'\| d\gamma(z, z') > \tau_{\text{drift}}$

When evaluating the overarching system, this translates to compounding non-IID properties. The system architecture must reconcile both the temporal shift occurring locally within node i , and the generalized non-IID distribution divergence occurring spatially between node i and node j , without ever aggregating the divergent sets [Ref].

Isolated Data Abstraction

A core assumption of this adaptation pipeline involves pre-processing strictness. To ensure that human-in-the-loop active learning does not violate constitutional surveillance mandates, the node never retains raw images. Edge nodes instantaneously collapse

raw camera feeds into abstract lower-dimensional bounding box coordinates or skeleton keypoint vectors. The algorithms presented henceforth evaluate solely these decoupled vector sequences $\mathbf{x} \in \mathbb{R}^k$.

LABEL-EFFICIENT DRIFT ADAPTATION FRAMEWORK

When drift shifts $P_{i^{(t+1)}}$ beyond predictable radii, classification networks must retrain. Given the strict labeling limits imposed by operators, the node must actively rank incoming vectors by their relative information gain and selectively query only those mapping to uncharted domain space.

Epistemic Uncertainty and Mutual Information

If we query an oracle exclusively based on simple Softmax entropy (e.g., $1 - \max p_i(y|x)$), the query pool becomes overwhelmingly saturated with "impossible" edge cases—such as abstract coordinates corrupted heavily by motion blur or extreme momentary occlusion (Aleatoric noise). Handing noisy garbage data to a human annotator wastes the annotation budget without improving the model's structural understanding of the new domain.

Instead, we employ Bayesian Active Learning by Disagreement (BALD). By placing a prior $p(w)$ over the internal weights, BALD seeks to query input vectors x which maximize the mutual information $\mathbb{H}\{y\}$ between predictions y and the hidden model parameters w :

$$\text{Equation: } \mathbb{H}\{y; w | x, \mathcal{D}_{\text{old}}\} = \mathbb{H}\{y|x, \mathcal{D}_{\text{old}}\} - \mathbb{E}_{w \sim p(w|\mathcal{D}_{\text{old}})}[\mathbb{H}\{y|x, w\}]$$

In this equation:

- The first term, $\mathbb{H}\{y|x\}$, is the overall predictive uncertainty of the distribution given the current data context.
- The second term, $\mathbb{E}[\mathbb{H}\{y\}]$, represents the expected confidence of the individual sampled network parameters.

Subtracting expected internal entropy from the global predictive entropy elegantly strips away Aleatoric noise. High mutual information signifies that individual parameter states are extremely confident in their predictions, but violently disagree with one another regarding the specific class. This disagreement pinpoints Epistemic Uncertainty—the precise boundary of the temporal distribution shift where the model explicitly lacks data structure.

Monte Carlo Dropout Integration

Exact Bayesian integration over modern deep network weight spaces is computationally intractable under edge-resource constraints. MC Dropout provides an operationally tractable approximation: by keeping dropout layers active (with dropout rate p_{drop}) at test time, the network produces stochastic posterior samples with no additional memory or computational burden on top of standard dropout training [Ref], [Ref].

For an input $\mathbf{x}_t$, the node is executed K times, producing independent stochastic forward passes, ultimately yielding a sample distribution $\left\{\hat{\mathbf{y}}^{(1)}, \dots, \hat{\mathbf{y}}^{(K)}\right\}$. Crucially, this distribution only provides approximate uncertainty guarantees - the dropout distribution is in fact a variational distribution that can underestimate the true epistemic uncertainty by virtue of being a poor approximation to the real posteriors (especially beyond the training manifold).

Algorithm \ref{alg:mc_bald} details how this directly translates into an on-device sample acquisition engine.

\begin{algorithm}[h]

```

\caption{Information-Theoretic Sample Acquisition (Edge)}
\begin{algorithmic}[1]
  \REQUIRE Abstract Inference Stream  $S$ , Pretrained Base Model  $M$ , Selection Threshold  $\tau_{\text{MI}}$ , Oracle Capacity  $C_{\text{max}}$ 
  \ENSURE Highly Informative Verified Batch  $B_{\text{train}}$ 
  \STATE  $B_{\text{train}} \leftarrow \emptyset$ , Count  $\leftarrow 0$ 
  \FOR{each generic feature vector  $x_t \in S$ }
    \STATE // Step 1: Stochastic MC Forward Passes
    \FOR{ $k=1$  to  $K=10$ }
      \STATE  $p_k \leftarrow M.\text{forward}(x_t, \text{dropout}=\text{True})$ 
    \ENDFOR
    \STATE // Step 2: Extract Expectation and Entropy
    \STATE  $p_{\text{mean}} \leftarrow \frac{1}{K} \sum_{k=1}^K p_k$ 
    \STATE  $H_{\text{total}} \leftarrow -\sum p_{\text{mean}} \log(p_{\text{mean}})$ 
    \STATE  $H_{\text{expected}} \leftarrow \frac{1}{K} \sum_{k=1}^K (-\sum p_k \log(p_k))$ 
    \STATE // Step 3: Compute Epistemic Disagreement
    \STATE  $MI \leftarrow H_{\text{total}} - H_{\text{expected}}$ 
    \IF{ $MI > \tau_{\text{MI}}$  \AND Count  $< C_{\text{max}}$ }
      \STATE // Model is fundamentally ignorant regarding  $x_t$ 
      \STATE  $y_{\text{verified}} \leftarrow \text{QueryHumanOracle}(x_t)$ 
      \STATE  $B_{\text{train}}.\text{add}(x_t, y_{\text{verified}})$ 
      \STATE Count  $\leftarrow$  Count + 1
    \ELSIF{ $H_{\text{total}} \leq \tau_{\text{confident}}$ }
      \STATE // Pseudo-label inclusion for base retention
      \STATE  $D_{\text{auto}}.\text{add}(x_t, \text{argmax}(p_{\text{mean}}))$ 
    \ENDIF
  \ENDFOR
  \RETURN  $B_{\text{train}} \cup D_{\text{auto}}$ 
\end{algorithmic}
\label{alg:mc_bald}
\end{algorithm}

```

Setting $K = 10$ proved optimal in terms of variance bounds while still allowing the active learning pipeline to avoid causing thermal throttling events in standard ARM Cortex cores during real-time edge streaming. Increasing K beyond 10 only provided marginal improvements in uncertainty estimation while introducing unacceptable latency spikes to the processing pipeline for sustained streaming workloads.

STABILITY CONSTRAINTS UNDER NOISY OBFUSCATION

Obtaining the highly optimized local dataset B_{train} is only the first step. To safely leverage these samples system-wide, the framework relies on downstream privacy-preserving federated aggregation. While this paper does not detail the extensive protocol infrastructure necessary for global routing or cryptographic privacy budgeting (see related structural definitions in [Ref]), it fundamentally depends on recognizing that the federation mechanism obfuscates incoming gradients by injecting Gaussian stochastic variance.

Mathematical Destabilization from Dimensionality

In privacy-preserving mechanisms, to guarantee mathematical bounds, local gradient vectors Δw_i are explicitly clipped to a specific norm bound C_{clip} , after which calibrated noise $\mathcal{N}(0, \sigma^2 C_{\text{clip}}^2 \mathbf{I}_d)$ is injected.

A critical observation is that the injected noise directly scales with the dimensionality d of the updated structural parameter vector. Active learning processes inherently yield small, sparse batches $\mathcal{D}_{\text{local}}$. The calculated true gradient $\nabla F(w)$ relies almost entirely on handfuls of highly impactful vectors.

If we unfreeze the full model space (e.g., $d \approx 11 \times 10^6$ parameters for ResNet-18), the cumulative sum of the stochastically independent Gaussian noise variables across the full array completely overwhelms the subtle, fragile gradient signals indicating the temporal environmental drift constraint. Under full-depth updating, the stationary radius collapses into generalized random-walk degradation:

Equation: $\text{Stationary Error} \propto \frac{\sigma}{\sqrt{d}} |B_{\text{train}}|$

Layer Freezing as Stability-Preservation Constraint

The prevailing view posits layer freezing exclusively as a communication optimization technique. Within an active edge-deployed setting subjected to privacy-noise perturbation, we formally elevate layer freezing to a stability-preservation principle.

Sparse active-learning gradients are inherently fragile—derived from handfuls of highly informative vectors. DP perturbation scales with the dimensionality d of the updated parameter vector. Unrestricted updates amplify stochastic noise across the full parameter space, overwhelming the subtle gradient signals indicating drift boundaries. Constrained trainable geometry preserves adaptation directionality.

By structurally immobilizing the early convolutional feature extractor and restricting optimization to the final three MLP classification layers, we reduce the trainable scope to $d_{\text{train}} \approx 5.5 \times 10^6$. This bounds the variance of injected noise matrices, ensuring that the fragile gradient signals harvested via epistemic mutual information are preserved through federated aggregation.

EMPIRICAL EVALUATION

We analyze the active learning drift compensation architecture through controlled simulations designed to rigorously emulate physical degradation.

Environment and Controlled Drift Profiles

The generalized assessment runs on a distributed array of five institutional edge nodes. In order to test adaptability without raising

ethical concerns, all active evaluations make use of heavily abstracted vector sets.

We constructed three severe temporal drift formulations spanning extended operational windows:

- Lighting Drift Scenario:** Simulates gradual sunset progressions and structural window degradation. Abstract vector spaces skewed universally by a sliding coordinate suppression factor (-15%) coupled with localized multi-variant Gaussian additive static.
- Demographic Turnover Scenario:** Replaces an advancing 15% cohort of the local sample arrays with extreme out-of-distribution abstract classifications.
- Spatial Occlusion Scenario:** Introduces fixed masking bounds across positional geometries, mirroring the accumulation of physical debris or shifted seating architectures typical in crowded dynamic spaces.

Metrics and Recovery Efficacy

To quantify active tracking success against silent deterioration, we compute the Drift Recovery Ratio (ρ). Let Δ_{acc} define the systemic total drop initialized by the silent drift trajectory relative to historical baseline, while Δ_{rec} measures the total accuracy regained executing the active adaptation protocols.

Equation: $\rho = \frac{\Delta_{\text{rec}}}{\Delta_{\text{acc}}}$

[FIGURE: Architectural Pipeline / Validation Curves]

Cross-node Temporal Recovery Curve. The unadapted model drops decisively to an 85.2% operational plateau. Assisted significantly by localized active pseudo-labeling boundaries, performance aggressively rebounds toward 93.0%.

Figure: `fig:accuracy_plot`

Table [ref{tab:drift}](#) details the specific breakdown of evaluation recoveries averaged across multiple evaluation runs.

TABLE: Drift Mitigation Comparative Matrix

Within the evaluated drift scenarios, the unadapted accuracy gap Δ_{acc} equates to 9.8pp (95.0% degrading to 85.2%). Uncertainty-guided adaptation recovered 7.8pp, returning operational stability to 93.0%. Tracking this yields $\rho = 7.8 / 9.8 \approx 0.796$. Reclaiming 79.6% of drift degradation under a 15% annotation budget is consistent with the hypothesis that selective epistemic acquisition substantially outperforms uniform sampling. We note that

simulated drift approximates but may not fully capture the compound complexity of organic environmental evolution.

Active Learning Pipeline Efficiency

A fundamental target of this work was minimizing operator intervention timelines. Without predictive targeting, a random sampling regimen mapped across a monthly drift cycle necessitated manually reviewing and verifying roughly 3,000 abstract records per discrete node location to obtain a robustly balanced dataset matching the environmental gradient footprint.

Conversely, the BALD Epistemic target array systematically narrowed the selection queue exclusively to challenging edge-case distributions. The simulation ultimately required isolated human interventions strictly approaching \approx 450 individual verification prompts. Achieving robust trajectory stabilization while reducing human workload budgets by \approx 85% establishes the core effectiveness of Monte Carlo acquisition metrics outperforming baseline randomness.

Ablation: Optimization Trajectory Stability

To empirically validate the mathematical necessity of the constrained parameter updating previously formalized in Section V, we executed ablation variations systematically unfurling the deep feature arrays to observe convergence reaction limits.

TABLE: Trajectory Variance vs. Freezing Geometry

Under the simulated environmental conditions, exposing 11-million dimensions to privacy-preserving aggregation noise overwhelmed the carefully curated active-learning labels, generating destructive feedback oscillations. Restricting modifications to the final 5.5M classification layers provided the stability constraint necessary to preserve the information signal from epistemic acquisition. The terminal-layer-only configuration (\approx 1M) achieved stability but suffered structural underfitting—insufficient capacity to capture the distributional shift. This confirms that layer freezing functions as a stability-preservation principle rather than merely a bandwidth optimization.

DISCUSSION

While the integration of Bayesian Active Learning provides an elegant countermeasure against extreme label sparsity on unmonitored edge stations, systemic limits demand recognition. Consistent long-term temporal adaptation naturally courts catastrophic environmental forgetting. Re-orienting the geometric classification bounds aggressively toward novel operational conditions eventually overwrites structurally critical historical representations. For instance, drifting optimally into distinct summer lighting formulations progressively erodes systemic competency involving heavy winter occlusion constraints.

Limitations

The active learning adaptation framework exposes several systemic limitations within the evaluated drift scenarios:

- **Catastrophic Forgetting:** Adapting to novel environmental conditions progressively erodes historical competence, necessitating replay buffers that may conflict with data-retention schedules.
- **Pseudo-Labeling Confirmation Bias:** Automatic labeling routines suffer from confirmation bias; highly confident but structurally incorrect representations reinforce errors rather than correcting them.
- **MC Dropout Approximation Limits:** MC Dropout provides tractable uncertainty approximation but does not represent a true posterior distribution. Uncertainty calibration may degrade in regions far from the training manifold.
- **Synthetic Drift Simplification:** Evaluation relied upon abstracted synthetic vectors and simulated drift trajectories, which

approximate but may not capture the full complexity of compound organic degradation.

- **Federated-Noise Sensitivity:** Despite layer-freezing constraints, high DP noise still degrades sparse gradient signals.
- **Annotation Fatigue:** The 15% annotation budget assumes consistent human oracle availability; delayed verification, intermittent connectivity, and operator fatigue may reduce effective annotation throughput.
- **No Long-Horizon Deployment Validation:** The simulated adaptation window does not validate multi-year drift trajectories or institutional personnel turnover effects on annotation quality.

CONCLUSION

This paper addressed the operational challenge of temporal distribution drift in edge-deployed neural networks under sparse-label constraints. Within the evaluated drift scenarios, measuring epistemic uncertainty via BALD-guided Monte Carlo Dropout curated focused retraining datasets locally, reducing human verification requirements to approximately 15% of the annotation burden required by uniform sampling.

The results are consistent with the hypothesis that selective uncertainty-guided acquisition substantially outperforms random sampling for drift adaptation. The framework recovered approximately 79.6% of drift-induced accuracy degradation under the simulated conditions. Layer freezing was validated not as a peripheral bandwidth optimization but as a stability-preservation constraint—restricting trainable geometry to prevent privacy-noise amplification from overwhelming fragile active-learning gradients.

Within the ScholarMaster architecture, this framework operates exclusively as the federated drift-compensation and uncertainty-guided adaptation subsystem—isolating epistemic uncertainty and stabilizing sparse-label adaptation without optimizing hierarchical aggregation (Paper 14), modifying raw perception pipelines, synthesizing deployment architecture (Paper 20), or governing runtime enforcement lifecycles (Paper 18).

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